

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400067
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Chen; Sheng-Hsiung et al.

Integrated circuit and method of forming same and a system

Abstract

A multi-bit flip-flop includes a first flip-flop and a second flip-flop. The first flip-flop has a first driving capability. The first flip-flop includes a first set pin configured to receive a first set signal. The second flip-flop has a second driving capability different from the first driving capability. The second flip-flop includes a second set pin configured to receive the first set signal, and the first set pin and the second set pin are coupled together. The first flip-flop and the second flip-flop are configured to share at least a first clock pin.

Inventors: Chen; Sheng-Hsiung (Hsinchu, TW), Chen; Wen-Hao (Hsinchu, TW), Ou; Hung-Chih (Hsinchu, TW), Ku; Chun-Yao (Hsinchu, TW), Wang; Shao-Huan (Hsinchu, TW)

Applicant: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

Family ID: 1000008782138

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

Appl. No.: 18/746888

Filed: June 18, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240338511 A1	Oct. 10, 2024

Related U.S. Application Data

continuation parent-doc US 18337245 20230619 US 12014131 child-doc US 18746888
continuation parent-doc US 17692767 20220311 US 11681853 20230620 child-doc US 18337245
continuation parent-doc US 17235262 20210420 US 11275886 20220315 child-doc US 17692767

Publication Classification

Int. Cl.: **G06F30/30** (20200101); **G06F30/3315** (20200101); **G06F30/337** (20200101);
G06F30/396 (20200101); **G06F30/398** (20200101); G06F30/392 (20200101);
G06F115/06 (20200101); G06F119/06 (20200101); G06F119/12 (20200101)

U.S. Cl.:

CPC **G06F30/398** (20200101); **G06F30/3315** (20200101); **G06F30/337** (20200101);
G06F30/396 (20200101); G06F30/392 (20200101); G06F2115/06 (20200101);
G06F2119/06 (20200101); G06F2119/12 (20200101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7260442	12/2006	Hwang et al.	N/A	N/A
9177096	12/2014	Sundareswaran et al.	N/A	N/A
9256709	12/2015	Yu et al.	N/A	N/A
9361417	12/2015	Arunachalam et al.	N/A	N/A
9513658	12/2015	Krishnamurthy et al.	N/A	N/A
9667230	12/2016	Fojtik et al.	N/A	N/A
9779197	12/2016	Wagner	N/A	H01L 27/0207
10338139	12/2018	Ge et al.	N/A	N/A
10641822	12/2019	Balasbrumanian et al.	N/A	N/A
10958252	12/2020	Wu et al.	N/A	N/A
10990745	12/2020	Chen et al.	N/A	N/A
11275886	12/2021	Chen et al.	N/A	N/A
11681853	12/2022	Chen	N/A	N/A
12014131	12/2023	Chen	N/A	G06F 30/398
2003/0014612	12/2002	Joy et al.	N/A	N/A
2014/0040838	12/2013	Liu et al.	N/A	N/A
2015/0227646	12/2014	Arunachalam et al.	N/A	N/A
2015/0278429	12/2014	Chang	N/A	N/A
2015/0318845	12/2014	Pasternak	N/A	N/A
2016/0241219	12/2015	Kim et al.	N/A	N/A
2016/0266604	12/2015	Krishnamurthy et al.	N/A	N/A
2017/0045576	12/2016	Balasubramanian et al.	N/A	N/A
2017/0063350	12/2016	Cheng et al.	N/A	N/A
2017/0292993	12/2016	Yoon et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105897221	12/2015	CN	N/A
108092660	12/2017	CN	N/A
2007173509	12/2006	JP	N/A
2013182600	12/2012	JP	N/A
201630352	12/2015	TW	N/A

OTHER PUBLICATIONS

R.P. Pokala et al., “Physical Synthesis for Performance Optimization,” Fifth Annual IEEE International ASIC Conference and Exhibit, Sep. 1992, pp. 34-37. DOI: 10.1109/ASIC.1992.270312 (Year: 1992). cited by applicant

S.-H. Wang et al., “Power-Driven Flip-Flop Merging and Relocation,” IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems, vol. 31, No. 2, Feb. 2012, pp. 180-191. (Year: 2012). cited by applicant

G. Barathi et al., “Lower Power High Speed Memory Architecture Using Multi-bit Flip-Flops,” Int’l Journal of Advanced Information Science and Technology (IJAIST), vol. 13, No. 13, May 2013, pp. 117-126. (Year: 2013). cited by applicant

C.-C. Lu et al., “Slack Budgeting and Slack to Length Converting for Multi-bit Flip-Flop Merging,” 2013 EDAA, IEEE, 6 pages. (Year: 2013). cited by applicant

S. Sharma et al., “Shift Register Design Using Two Bit Flip-Flop,” Proc. of 2014 RAECS UIET Panjab University Chandigarh, IEEE, 5 pages. (Year: 2014). cited by applicant

A. Abinaya et al., “Efficient Flip-Flop Merging Technique for Clock Power Reduction,” IEEE Sponsored 2nd Int’l Conference on Electronics and Communication System (ICECS 2015), pp. 326-330. (Year: 2015). cited by applicant

D. Gluzer et al., “Probability-Driven Multibit Flip-Flop Integration With Clock Gating,” IEEE Trans. on Very Large Scale Integration (VLSI) Systems, vol. 25, No. 3, Mar. 2017, pp. 1173-1177. (Year 2017). cited by applicant

Office Action dated Jan. 7, 2022 for corresponding case No. TW 11120031400. (pp. 1-4). cited by applicant

Xunji, Zheng, “Research on Digital Circuits Low Power Design Method Based on Multi Bit Flip Flop”, A thesis submitted to Xidian University; May 2017, 77 pages. cited by applicant

Xinjie, Wang, “SoC Low Power Design and Implementation Based on Multi-Bit Flip Flop”, A thesis submitted to Graduate School of National University of Defense Technology Changsha, Hunan, P.R. China, Mar. 2015, 87 pages. cited by applicant

Santos, Cristiano et al., “Multi-Bit Flip-Flop Usage Impact on Physical Synthesis”, State Intellectual Property Office of China, Downloaded on Mar. 18, 2024, IEEE Xplore, 6 pages. cited by applicant

“Research on Low Power Technology Based on Multi-Bit Flip Flop” 71 pages. cited by applicant

Primary Examiner: Garbowski; Leigh M

Attorney, Agent or Firm: Hauptman Ham, LLP

Background/Summary

PRIORITY CLAIM (1) This application is a continuation of U.S. application Ser. No. 18/337,245, filed Jun. 19, 2023, now U.S. Pat. No. 12,014,131, issued Jun. 18, 2024, which is a continuation of U.S. application Ser. No. 17/692,767, filed Mar. 11, 2022, now U.S. Pat. No. 11,681,853, issued Jun. 20, 2023, which is a continuation of U.S. application Ser. No. 17/235,262, filed Apr. 20, 2021, now U.S. Pat. No. 11,275,886, issued Mar. 15, 2022, which is a continuation of U.S. application Ser. No. 16/559,534, filed Sep. 3, 2019, now U.S. Pat. No. 10,990,745, issued Apr. 27, 2021, which claims the benefit of U.S. Provisional Application No. 62/733,878, filed Sep. 20, 2018, which are herein incorporated by reference in their entireties.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has produced a wide variety of digital devices to address issues in a number of different areas. Some of these digital devices, such as level shifter circuits, are configured to enable operation of circuits capable of operation in different voltage domains. As ICs have become smaller and more complex, operating voltages of these digital devices continue to decrease affecting IC performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a schematic diagram of a multi-bit flip-flop, in accordance with some embodiments.

(3) FIG. 2A is a circuit diagram of a circuit, in accordance with some embodiments.

(4) FIG. 2B is a circuit diagram of a circuit, in accordance with some embodiments.

(5) FIG. 3 is a functional flowchart of at least a portion of an IC design and manufacturing flow, in accordance with some embodiments.

(6) FIG. 4 is a flowchart of a method, in accordance with some embodiments.

(7) FIGS. 5A-5B are schematic views of layout diagrams of flip-flops before and after modifications, in accordance with some embodiments.

(8) FIGS. 6A-6D are schematic views of layout diagrams of flip-flops before and after modifications, in accordance with some embodiments.

(9) FIG. 7 is a schematic view of a system for designing an IC layout design, in accordance with some embodiments.

(10) FIG. 8 is a block diagram of an integrated circuit (IC) manufacturing system, and an IC manufacturing flow associated therewith, in accordance with at least one embodiment of the present disclosure.

DETAILED DESCRIPTION

(11) The following disclosure provides different embodiments, or examples, for implementing features of the provided subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of

simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) In accordance with some embodiments, an integrated circuit includes a first flip-flop and a second flip-flop. The first flip-flop has a first driving current capability. The second flip-flop has a second driving current capability different from the first driving current capability. The first flip-flop and the second flip-flop are part of a multibit flip-flop (MBFF) configured to share at least a first clock pin. The first clock pin is configured to receive the first clock signal. In some embodiments, by configuring the first flip-flop and the second flip-flop with different driving current capabilities, the MBFF is a mixed driving multi-bit flip-flop.

(14) In some embodiments, the first driving current capability of the first flip-flop is based on a first number of fins in one or more transistors in the first flip-flop. In some embodiments, the second driving current capability of the second flip-flop is based on a second number of fins in one or more transistors in the second flip-flop. In some embodiments, the second number of fins is different from the first number of fins.

(15) In some embodiments, the first driving current capability of the first flip-flop is based on a first threshold voltage of one or more transistors in the first flip-flop. In some embodiments, the second driving current capability of the second flip-flop is based on a second threshold voltage of one or more transistors in the second flip-flop. In some embodiments, the second threshold voltage is different from the first threshold voltage.

(16) In some embodiments, by configuring the MBFF as a mixed driving multi-bit flip-flop, a number of duplicate inverters in the clock path of the MBFF are reduced resulting in the MBFF having less input pins for a corresponding clock signal, and the MBFF occupies less area compared with other approaches.

(17) In some embodiments, by configuring the MBFF as a mixed driving multi-bit flip-flop, a number of duplicate inverters in the clock path of the MBFF are reduced resulting in a lower total clock dynamic power consumption compared with other approaches. In some embodiments, by configuring the MBFF as a mixed driving multi-bit flip-flop, the power consumption of each flip-flop in the MBFF is optimized compared with other approaches.

(18) Furthermore, in some embodiments, by configuring the MBFF as a mixed driving multi-bit flip-flop, the power consumption of each flip-flop in the MBFF is optimized as flip-flops that may be debanked in the design flow order to fix timing violations in other approaches, are upsized or downsized in the design flow within the same MBFF rather than being debanked to another flip-flop as in the other approaches.

Multi-Bit Flip-Flop

(19) FIG. 1 is a schematic diagram of a multi-bit flip-flop (MBFF) **100**, in accordance with some embodiments.

(20) MBFF **100** comprises a flip-flop **102**, a flip-flop **104**, a flip-flop **106**, a flip-flop **108**, an inverter **120**, an inverter **122** and a clock input pin **130**. MBFF **100** is a four bit flip-flop. In other words, MBFF includes four flip-flops (e.g., flip-flops **102**, **104**, **106** and **108**). Other numbers of bits or corresponding flip-flops in MBFF **100** are within the scope of the present disclosure. In some embodiments, MBFF **100** is part of an integrated circuit (not shown) that includes other MBFF's, similar to MBFF **100**, or one or more other flip-flops.

(21) MBFF **100** is configured to receive input signals D1, D2, D3 and D4, and to receive clock

signal CLK on the clock input pin **130**. MBFF **100** is configured to generate output signals Q1, Q2, Q3 and Q4.

(22) Flip-flops **102**, **104**, **106** and **108** are configured to receive corresponding input signals D1, D2, D3 and D4 on corresponding input terminals (not labelled). Flip-flops **102**, **104**, **106** and **108** are configured to generate corresponding output signals Q1, Q2, Q3 and Q4, and to output the corresponding output signals Q1, Q2, Q3 and Q4 on corresponding output terminals (not labelled).

(23) Each of flip-flops **102**, **104**, **106** and **108** is further configured (not shown) to receive clock signal CLK and inverted clock signal CLKB. Each of flip-flops **102**, **104**, **106** and **108** is coupled to inverters **120** and **122**. In some embodiments, each of flip-flops **102**, **104**, **106** and **108** is configured (not shown) to share input pin **130**. Each of flip-flops **102**, **104**, **106** and **108** is further configured to receive clock signal CLK from input pin **130**, and is configured to receive inverted clock signal CLKB from inverter **120**. In some embodiments, each of flip-flops **102**, **104**, **106** and **108** are configured to receive clock signal CLK' from inverter **122**. In some embodiments, clock signal CLK' is a buffered version of clock signal CLK. In some embodiments, inverted clock signal CLKB is inverted from the clock signal CLK.

(24) In some embodiments, one or more of flip-flops **102**, **104**, **106** and **108** are edge triggered flip-flops. In some embodiments, one or more of flip-flops **102**, **104**, **106** and **108** includes a DQ flip-flop, an SR-flip-flop, a T flip-flop, a JK flip-flop, or the like. Other types of flip-flops or configurations for at least flip-flop **102**, **104**, **106** or **108** are within the scope of the present disclosure.

(25) An input terminal of inverter **120** is coupled to the clock input pin **130**, and is configured to receive clock signal CLK. An output terminal of inverter **120** is coupled to an input terminal of inverter **122** and is configured to output inverted clock signal CLKB.

(26) An input terminal of inverter **122** is configured to receive inverted clock signal CLKB. An output terminal of inverter **120** is configured to output clock signal CLK'. Other configurations for at least inverter **120** or **122** are within the scope of the present disclosure.

(27) MBFF **100** is configured as a mixed driving multi-bit flip-flop. For example, MBFF **100** includes flip-flops configured with at least two different driving current capabilities. In some embodiments, each of the flip-flops contained in MBFF **100** are configured to have different driving current capability. Other numbers of different driving current capabilities for MBFF **100** are within the scope of the present disclosure. For example, in some embodiments, MBFF **100** includes three different flip-flops, each of the three different flip-flops is configured with a different driving current capability from the other.

(28) Flip-flop **102** and flip-flop **104** (collectively referred to as "a set of flip-flops **110**") are each configured to have a first driving current capability FF1. In some embodiments, the first driving current capability FF1 corresponds to the driving current conducted by at least flip-flop **102** or flip-flop **104**.

(29) Flip-flop **106** and flip-flop **108** (collectively referred to as "a set of flip-flops **112**") are each configured to have a second driving current capability FF0. In some embodiments, the second driving current capability FF0 corresponds to the driving current conducted by at least flip-flop **106** or flip-flop **108**. In some embodiments, the second driving current capability FF0 is different from the first driving current capability FF1.

(30) In some embodiments, the first driving current capability FF1 of at least flip-flop **102** or flip-flop **104** is based on a first number of fins F1 in one or more transistors in flip-flop **102** or flip-flop **104**.

(31) In some embodiments, the second driving current capability FF0 of at least flip-flop **106** or flip-flop **108** is based on a second number of fins F2 in one or more transistors in flip-flop **106** or flip-flop **108**. In some embodiments, the first number of fins F1 is different from the second number of fins F2. In some embodiments, at least the first number of fins F1 or the second number of fins F2 includes a single fin. In some embodiments, at least the first number of fins F1 or the

second number of fins F2 includes multiple fins. In some embodiments, the use of multiple fins increases channel width and current driving strength of the transistors in the set of flip-flops **110** or **112**. In some embodiments, additional fins are used to add further current driving strength, and these arrangements are within the scope of the present disclosure.

(32) In some embodiments, the first number of fins F1 and the first driving current capability FF1 have a direct relationship. For example, in some embodiments, as the first number of fins F1 in one or more transistors in flip-flop **102** or flip-flop **104** is increased, the first driving current capability FF1 is also increased. For example, in some embodiments, as the first number of fins F1 in one or more transistors in flip-flop **102** or flip-flop **104** is decreased, the first driving current capability FF1 is also decreased.

(33) In some embodiments, the second number of fins F2 and the second driving current capability FF0 have a direct relationship. For example, in some embodiments, as the second number of fins F2 in one or more transistors in flip-flop **106** or flip-flop **108** is increased, the second driving current capability FF0 is also increased. For example, in some embodiments, as the second number of fins F2 in one or more transistors in flip-flop **106** or flip-flop **108** is decreased, the second driving current capability FF0 is also decreased.

(34) Thus, by adjusting the first number of fins F1 in at least flip-flop **102** or flip-flop **104** to be different from the second number of fins F2 in at least flip-flop **106** or flip-flop **108**, the first driving current capability FF1 and the second current driving capability FF0 are also adjusted.

(35) In some embodiments, the first driving current capability FF1 of at least flip-flop **102** or flip-flop **104** is based on a first threshold voltage Vth1 of one or more transistors in flip-flop **102** or flip-flop **104**.

(36) In some embodiments, the second driving current capability FF0 of at least flip-flop **106** or flip-flop **108** is based on a second threshold voltage Vth2 of one or more transistors in flip-flop **106** or flip-flop **108**. In some embodiments, the first threshold voltage Vth1 is different from the second threshold voltage Vth2. In some embodiments, the first threshold voltage Vth1 is at least a high threshold voltage (HVT), a low threshold voltage (LVT) or a standard threshold voltage (SVT). In some embodiments, HVT is greater than at least LVT or SVT. In some embodiments, SVT is greater than LVT. In some embodiments, the second threshold voltage Vth2 is at least HVT, LVT or SVT.

(37) In some embodiments, the first threshold voltage Vth1 and the first driving current capability FF1 have an inverse relationship. For example, in some embodiments, as the first threshold voltage Vth1 of one or more transistors in flip-flop **102** or flip-flop **104** is increased, the first driving current capability FF1 is decreased. For example, in some embodiments, as the first threshold voltage Vth1 of one or more transistors in flip-flop **102** or flip-flop **104** is decreased, the first driving current capability FF1 is increased.

(38) In some embodiments, the second threshold voltage Vth2 and the second driving current capability FF0 have an inverse relationship. For example, in some embodiments, as the second threshold voltage Vth2 of one or more transistors in flip-flop **106** or flip-flop **108** is increased, the second driving current capability FF0 is decreased. For example, in some embodiments, as the second threshold voltage Vth2 of one or more transistors in flip-flop **106** or flip-flop **108** is decreased, the second driving current capability FF0 is increased.

(39) In some embodiments, the first threshold voltage Vth1 is dependent upon at least an oxide thickness or an oxide material of the transistors in the set of flip-flops **110**, a doping profile of the transistors in the set of flip-flops **110**, a transistor device geometry including a channel width of the transistors in the set of flip-flops **110**, a transistor size of the transistors in the set of flip-flops **110**, or the like.

(40) In some embodiments, the second threshold voltage Vth2 is dependent upon at least an oxide thickness or an oxide material of the transistors in the set of flip-flops **112**, a doping profile of the transistors in the set of flip-flops **112**, a transistor device geometry including a channel width of the

transistors in the set of flip-flops **112**, a transistor size of the transistors in the set of flip-flops **112**, or the like.

(41) In some embodiments, by configuring MBFF **100** as a mixed driving multi-bit flip-flop, a number of duplicate inverters in the clock path of MBFF **100** are reduced resulting in MBFF **100** having less input pins for a corresponding clock signal, and MBFF **100** occupying less area compared with other approaches.

(42) In some embodiments, by configuring MBFF **100** as a mixed driving multi-bit flip-flop, a number of duplicate inverters in the clock path of MBFF **100** are reduced resulting in a lower total clock dynamic power consumption compared with other approaches. In some embodiments, by configuring MBFF **100** as a mixed driving multi-bit flip-flop, the power consumption of each flip-flop in MBFF **100** is optimized compared with other approaches.

(43) Furthermore, in some embodiments, by configuring MBFF **100** as a mixed driving multi-bit flip-flop, the power consumption of each flip-flop in MBFF **100** is optimized as flip-flops that may be debanked in order to fix timing violations in other approaches, are upsized or downsized within the same MBFF rather than being debanked to another flip-flop as in the other approaches.

(44) In other words, in some embodiments, MBFF **100** can maintain the benefits of being part of a multi-bit flip-flop without the timing violations or the extra power consumption of other approaches.

(45) FIG. **2A** is a circuit diagram of a circuit **200A**, in accordance with some embodiments.

(46) Circuit **200A** is an embodiment of MBFF **100** of FIG. **1**. In some embodiments, circuit **200A** or circuit **200B** (FIG. **2B**) is an MBFF circuit. In some embodiments, circuit **200A** or circuit **200B** are part of an integrated circuit.

(47) Circuit **200A** comprises a flip-flop **202**, a flip-flop **204**, a flip-flop **206**, a flip-flop **208**, a clock input pin **230**, a set input pin **232** and a reset input pin **234**.

(48) Circuit **200A** is a four bit flip-flop, and each bit is associated with a corresponding flip-flop (e.g., flip-flops **202**, **204**, **206** and **208**). In other words, circuit **200A** includes four flip-flops (e.g., flip-flops **202**, **204**, **206** and **208**). Other numbers of bits or numbers of corresponding flip-flops in circuit **200A** are within the scope of the present disclosure. In some embodiments, circuit **200A** is part of an integrated circuit (not shown) that includes other MBFFs, similar to MBFF **100**, or one or more other flip-flops.

(49) Flip-flop **202** and flip-flop **204** (collectively referred to as “a set of flip-flops **210**”) are each configured to have the first driving current capability FF1. In some embodiments, the first driving current capability FF1 corresponds to the driving current conducted by at least flip-flop **202** or flip-flop **204**.

(50) Flip-flop **206** and flip-flop **208** (collectively referred to as “a set of flip-flops **212**”) are each configured to have the second driving current capability FF0. In some embodiments, the second driving current capability FF0 corresponds to the driving current conducted by at least flip-flop **206** or flip-flop **208**.

(51) Flip-flops **202**, **204**, **206** and **208** are embodiments of corresponding flip-flops **102**, **104**, **106** and **108** of FIG. **1**, and similar detailed description is omitted. Set of flip-flops **210** and **212** are embodiments of corresponding set of flip-flops **110** and **112** of FIG. **1**, and similar detailed description is omitted. Clock input pin **230** is an embodiment of clock input pin **130** of FIG. **1**, and similar detailed description is omitted.

(52) Each of flip-flops **202**, **204**, **206** and **208** are a DQ flip-flop. In some embodiments, one or more of flip-flops **202**, **204**, **206** and **208** includes an SR-flip-flop, a T flip-flop, a JK flip-flop, or the like. Other types of flip-flops or configurations for at least flip-flop **202**, **204**, **206** and **208** are within the scope of the present disclosure.

(53) Each of flip-flops **202**, **204**, **206** and **208** has a corresponding clock input terminal configured to receive clock signal CLK, a corresponding set input terminal configured to receive a set signal S1, and a corresponding reset input terminal configured to receive a reset signal R1.

(54) In some embodiments, each of flip-flops **202**, **204**, **206** and **208** is configured to share the clock input pin **230**, a set input pin **232** and a reset input pin **234**. Each of flip-flops **202**, **204**, **206** and **208** is configured to receive the clock signal CLK from input pin **230**, is configured to receive the set signal S1 from the set input pin **232**, and is configured to receive the reset signal R1 from reset input pin **234**. In some embodiments, the set input terminals of flip-flops **202**, **204**, **206** and **208** are coupled together and configured to receive the set signal S1 from the set input pin **232**. In some embodiments, the reset input terminals of flip-flops **202**, **204**, **206** and **208** are coupled together and configured to receive the reset signal R1 from the set input pin **234**.

(55) FIG. 2B is a circuit diagram of a circuit **200B**, in accordance with some embodiments.

(56) Circuit **200B** is an embodiment of circuit **100** of FIG. 1.

(57) Circuit **200B** is a variation of circuit **200A**, and similar detailed description is therefore omitted. For example, circuit **200B** illustrates an example of where circuit **200B** includes a 2 bit, scan chain multi-bit flip-flop.

(58) Components that are the same or similar to those in one or more of FIGS. 1, 2A-2B, 5A-5B and 6A-6D (shown below) are given the same reference numbers, and detailed description thereof is thus omitted.

(59) Circuit **200B** comprises a flip-flop **250**, a flip-flop **252**, a multiplexer **260**, a multiplexer **262**, a clock input pin **230'**, a set input pin **232'**, a reset input pin **234'**, a scan enable pin **270** and a scan in pin **272**.

(60) Circuit **200B** is a 2 bit MBFF arranged in a scan chain configuration. Each bit in circuit **200B** is associated with a corresponding flip-flop (e.g., flip-flops **250** and **252**). In other words, circuit **200B** includes two flip-flops (e.g., flip-flops **250** and **252**). Other numbers of bits or numbers of corresponding flip-flops in circuit **200B** are within the scope of the present disclosure. In some embodiments, circuit **200B** is part of an integrated circuit (not shown) that includes other MBFFs, similar to MBFF **100**, or one or more other flip-flops.

(61) Flip-flop **250** is an embodiment of flip-flop **102** or **106** of FIG. 1, flip-flop **252** is an embodiment of flip-flop **104** or **108** of FIG. 1, and similar detailed description is omitted. Clock input pin **230'** is an embodiment of clock input pin **130** of FIG. 1 or clock input pin **230** of FIG. 2A, set input pin **232'** is an embodiment of set input pin **232** of FIG. 2A, reset input pin **234'** is an embodiment of reset input pin **234** of FIG. 2A, and similar detailed description is omitted.

(62) Flip-flop **250** and flip-flop **252** are each configured to have the first driving current capability FF1 or the second driving current capability FF0.

(63) Each of flip-flops **250** or **252** are a DQ flip-flop. In some embodiments, one or more of flip-flops **250** or **252** includes an SR-flip-flop, a T flip-flop, a JK flip-flop, or the like. Other types of flip-flops or configurations for at least flip-flop **250** or **252** are within the scope of the present disclosure.

(64) Each of flip-flops **250** or **252** has a corresponding clock input terminal configured to receive clock signal CLK, a corresponding set input terminal configured to receive a set signal S1, and a corresponding reset input terminal configured to receive a reset signal R1.

(65) In some embodiments, each of flip-flops **250** and **252** is configured to share the clock input pin **230'**, a set input pin **232'** and a reset input pin **234'**. Each of flip-flops **250** and **252** is configured to receive the clock signal CLK from input pin **230'**, is configured to receive the set signal S1 from the set input pin **232'**, and is configured to receive the reset signal R1 from reset input pin **234'**. In some embodiments, the set input terminals of flip-flops **250** and **252** are coupled together and configured to receive the set signal S1 from the set input pin **232'**. In some embodiments, the reset input terminals of flip-flops **250** and **252** are coupled together and configured to receive the reset signal R1 from the set input pin **234'**.

(66) Multiplexer **260** has a first input terminal configured to receive input signal D1 or D3, a second input terminal configured to receive scan in signal SI1, a third input terminal configured to receive the scan enable signal SE1, and an output terminal configured to output the input signal of

flip-flop **250** (e.g., signal **D1**, **D3** or **SI1**).

(67) Multiplexer **262** has a first input terminal configured to receive input signal **D2** or **D4**, a second input terminal configured to receive output signal **Q1** from flip-flop **250**, a third input terminal configured to receive the scan enable signal **SE1**, and an output terminal configured to output the input signal of flip-flop **252** (e.g., signal **D2**, **D4** or **Q1**).

(68) In some embodiments, multiplexers **260** and **262** are coupled to and configured to share the scan enable input pin **270**. In some embodiments, the third terminals of multiplexer **260** and **262** are coupled together and configured to receive the scan enable signal **SE1** from the scan enable input pin **270**.

(69) The first input terminal of multiplexer **260** is configured to receive input signal **D1** or **D3**, and the first input terminal of multiplexer **262** is configured to receive input signal **D2** or **D4** based on whether flip-flop **250** corresponds to an embodiment of flip-flop **102** or **106** of FIG. **1**, and whether flip-flop **252** corresponds to an embodiment of flip-flop **104** or **108** of FIG. **1**. For example, if flip-flop **250** is an embodiment of flip-flop **102**, then the first input terminal of multiplexer **260** is configured to receive input signal **D1**, and if flip-flop **252** is an embodiment of flip-flop **104**, then the first input terminal of multiplexer **262** is configured to receive input signal **D2**. For example, if flip-flop **250** is an embodiment of flip-flop **106**, then the first input terminal of multiplexer **260** is configured to receive input signal **D3**, and if flip-flop **252** is an embodiment of flip-flop **108**, then the first input terminal of multiplexer **262** is configured to receive input signal **D4**.

(70) FIG. **3** is a functional flow chart of at least a portion of an IC design and manufacturing flow **300**, in accordance with some embodiments. The design and manufacturing flow **300** utilizes one or more electronic design automation (EDA) tools for generating, optimizing and/or verifying a design of an IC before manufacturing the IC. The EDA tools, in some embodiments, are one or more sets of executable instructions for execution by a processor or controller or a programmed computer to perform the indicated functionality. In at least one embodiment, the IC design and manufacturing flow **300** is performed by a design house of an IC manufacturing system discussed herein with respect to FIG. **3**.

(71) At IC design operation **310**, a design of an IC is provided by a circuit designer. In some embodiments, the design of the IC comprises an IC schematic, i.e., an electrical diagram, of the IC. In some embodiments, the schematic is generated or provided in the form of a schematic netlist, such as a Simulation Program with Integrated Circuit Emphasis (SPICE) netlist. Other data formats for describing the design are usable in some embodiments. In some embodiments, a pre-layout simulation is performed on the design to determine whether the design meets a predetermined specification. When the design does not meet the predetermined specification, the IC is redesigned. In at least one embodiment, a pre-layout simulation is omitted from FIG. **3**. In at least one embodiment, method **300** further includes a pre-layout simulation performed after operation **310**.

(72) At Automatic Placement and Routing (APR) operation **320**, a layout diagram of the IC is generated based on the IC schematic. The IC layout diagram comprises the physical positions of various circuit elements of the IC as well as the physical positions of various nets interconnecting the circuit elements. For example, the IC layout diagram is generated in the form of a Graphic Design System (GDS) or GDSII file. Other data formats for describing the design of the IC are within the scope of various embodiments. In the example configuration in FIG. **3**, the IC layout diagram is generated by an EDA tool, such as an APR tool. The APR tool receives the design of the IC in the form of a netlist as described herein. The APR tool performs floor planning to identify circuit elements, which are to be electrically connected to each other and which are to be placed in close proximity to each other, for reducing the area of the IC and/or reducing time delays of signals travelling over the interconnections or nets connecting the electrically connected circuit elements. In some embodiments, the APR tool performs partitioning to divide the design of the IC into a plurality of blocks or groups, such as clock and logic groups. In the example configuration in FIG. **3**, the APR tool performs one or more of a power planning operation **322**, a cell placement

operation **324**, a clock tree synthesis (CTS) operation **326** or a routing operation **328**.

(73) At power planning operation **322**, the APR tool performs power planning, based on the partitioning and/or the floor planning of the IC design, to generate a power grid structure which includes several conductive layers, such as metal layers. In some embodiments, one metal layer of the metal layers includes power lines or power rails extending in one direction, e.g., horizontally in a plan view. In some embodiments, another metal layer of the metal layers includes power lines or power rails extending in an orthogonal direction, e.g., vertically in a plan view.

(74) At cell placement operation **324**, the APR tool performs cell placement. For example, standard cells (also referred to herein as “cells”) configured to provide pre-defined functions and having pre-designed layout diagrams are stored in one or more cell libraries **325**. The APR tool accesses various cells from one or more cell libraries **325**, and places the cells in an abutting manner to generate an IC layout diagram corresponding to the IC schematic.

(75) The generated IC layout diagram includes the power grid structure and a plurality of cells, each cell including one or more circuit elements and/or one or more nets. In some embodiments, each cell includes one or more flip-flops or one or more multi-bit flip-flops, similar to the multi-bit flip-flop **100** of FIG. **1**, the multi-bit flip-flops **200A** of FIG. **2A** and the multi-bit flip-flop **200B** of FIG. **2B**. In some embodiments, a cell includes a logic gate cell. In some embodiments, a logic gate cell includes an AND, OR, NAND, NOR, XOR, INV, AND-OR-Invert (AOI), OR-AND-Invert (OAI), MUX, Flip-flop, BUFF, Latch, delay, clock cells, or the like. In some embodiments, a cell includes a memory cell. In some embodiments, a memory cell includes a static random access memory (SRAM), a dynamic RAM (DRAM), a resistive RAM (RRAM), a magnetoresistive RAM (MRAM), a read only memory (ROM), or the like. In some embodiments, a circuit element is an active element or a passive element. Examples of active elements include, but are not limited to, transistors and diodes. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), or the like, FinFETs, planar MOS transistors with raised source/drains, or the like. Examples of passive elements include, but are not limited to, capacitors, inductors, fuses, and resistors. Examples of nets include, but are not limited to, vias, conductive pads, conductive traces, and conductive redistribution layers, or the like.

(76) At clock tree synthesis (CTS) operation **326**, the APR tool performs CTS to minimize skew and/or delays potentially present due to the placement of circuit elements in the IC layout diagram. The CTS includes an optimization process to ensure that signals are transmitted and/or arrived at appropriate timings. For example, during the optimization process within the CTS, one or more buffers are inserted into the IC layout diagram to add and/or remove slack (timing for signal arrival) to achieve a desired timing. In some embodiments, operation **326** includes performing a timing analysis (similar to operation **404** of method **400** (shown in FIG. **4**)) of one or more critical paths that include one or more multi-bit flip-flops to determine timing violations in the one or more critical paths. The described CTS of operation **326** is an example. Other arrangements or operations are within the scope of various embodiments. For example, in one or more embodiments, one or more of the described operations are repeated or omitted.

(77) At routing operation **328**, the APR tool performs routing to route various nets interconnecting the placed circuit elements. The routing is performed to ensure that the routed interconnections or nets satisfy a set of constraints. For example, routing operation **328** includes global routing, track assignment and detailed routing. During the global routing, routing resources used for interconnections or nets are allocated. For example, the routing area is divided into a number of sub-areas, pins of the placed circuit elements are mapped to the sub-areas, and nets are constructed as sets of sub-areas in which interconnections are physically routable. During the track assignment, the APR tool assigns interconnections or nets to corresponding conductive layers of the IC layout

diagram. During the detailed routing, the APR tool routes interconnections or nets in the assigned conductive layers and within the global routing resources. For example, detailed, physical interconnections are generated within the corresponding sets of sub-areas defined at the global routing and in the conductive layers defined at the track assignment. After routing operation **328**, the APR tool outputs the IC layout diagram including the power grid structure, placed circuit elements and routed nets. The described APR tool is an example. Other arrangements are within the scope of various embodiments. For example, in one or more embodiments, one or more of the described operations are omitted.

(78) At sign-off operation **330**, one or more physical and/or timing verifications are performed. For example, sign-off operation **330** includes one or more of a resistance and capacitance (RC) extraction, a layout-versus-schematic (LVS) check, a design rule check (DRC) or a timing sign-off check (also referred to as a post-layout simulation). In some embodiments, other verification processes are performed.

(79) In some embodiments, an RC extraction is performed, e.g., by an EDA tool, to determine parasitic parameters, e.g., parasitic resistance and parasitic capacitance, of components in the IC layout diagram for timing simulations in a subsequent operation.

(80) In some embodiments, an LVS check is performed to ensure that the generated IC layout diagram corresponds to the design of the IC. Specifically, an LVS checking tool, i.e., an EDA tool, recognizes electrical components as well as connections in the space between the patterns of the generated IC layout diagram. The LVS checking tool then generates a layout netlist representing the recognized electrical components and connections. The layout netlist generated from the IC layout diagram is compared, by the LVS checking tool, with the schematic netlist of the design of the IC. If the two netlists match within a matching tolerance, the LVS check is passed. Otherwise, correction is made to at least one of the IC layout diagram or the design of the IC by returning the process to IC design operation **310** and/or APR operation **320**.

(81) In some embodiments, a DRC is performed, e.g., by an EDA tool, to ensure that the IC layout diagram satisfies certain manufacturing design rules, i.e., to ensure manufacturability of the IC. If one or more design rules is/are violated, correction is made to at least one of the IC layout diagram or the design of the IC by returning the process to IC design operation **310** and/or APR operation **320**. Examples of design rules include, but are not limited to, a width rule which specifies a minimum width of a pattern in the IC layout diagram, a spacing rule which specifies a minimum spacing between adjacent patterns in the IC layout diagram, an area rule which specifies a minimum area of a pattern in the IC layout diagram, or the like.

(82) In some embodiments, a timing sign-off check (post-layout simulation) is performed, e.g., by an EDA tool, to determine, taking the extracted parasitic parameters into account, whether the IC layout diagram meets a predetermined specification of one or more timing requirements. If the simulation indicates that the IC layout diagram does not meet the predetermined specification, e.g., if the parasitic parameters cause undesirable delays, correction is made to at least one of the IC layout diagram or the design of the IC by returning the process to IC design operation **310** and/or APR operation **320**. Otherwise, the IC layout diagram is passed to manufacture or additional verification processes.

(83) In operation **340**, the integrated circuit is manufactured based on the layout diagram. In some embodiments, the layout diagram of method **300** includes layout diagram **600B** or layout diagram **600D** which are usable to manufacture the integrated circuit, such as one or more of MBFF **100**, circuit **200A** of FIG. 2A or circuit **200B** of FIG. 2B. In some embodiments, the integrated circuit manufactured by operation **340** includes at least MBFF **100**, circuit **200A** of FIG. 2A or circuit **200B** of FIG. 2B. In some embodiments, operation **340** of method **300** comprises manufacturing at least one mask based on the layout diagram, and manufacturing the integrated circuit based on the at least one mask. In some embodiments, operation **340** is performed by IC manufacturing system **800** (FIG. 8). In some embodiments, one or more of the above-described operations are omitted.

(84) An additional verification process involves a multi-bit flip-flop (MBFF) power analysis **350**. In some embodiments, the MBFF power analysis **350** is performed by the APR tool or by another EDA tool. In some embodiments, the MBFF power analysis **350** is performed to determine the power consumed by at least one flip-flop in the multi-bit flip-flop, such as MBFF **100**, or at least one flip-flop in an integrated circuit, e.g., circuit **200A-200B** of FIGS. **2A-2B**. In some embodiments, the MBFF power analysis **350** is also configured to perform a timing analysis of at least one flip-flop in the multi-bit flip-flop, such as MBFF **100**, or at least one flip-flop in an integrated circuit, e.g., circuit **200A-200B** of FIGS. **2A-2B**. In some embodiments, the MBFF power analysis **350** is configured to optimize the power consumed by one or more flip-flops, such as MBFF **100** or circuit **200A** or **200B** of FIG. **2A** or **2B**, while also passing a timing analysis of the flip-flops in the integrated circuit. In some embodiments, portions of method **400** of FIG. **4** are an embodiment of at least the MBFF power analysis **350**.

(85) In some embodiments, MBFF power analysis **350** is part of a banking engine or a de-banking engine. In some embodiments, a banking engine is a portion of APR **320** that is configured to bank or group a plurality of flip-flops into one or more multibit flip-flops (MBFFs), as implemented by operation **402** of method **400** or shown by FIGS. **5A-5B**. In some embodiments, a de-banking engine is a portion of APR **320** that is configured to de-bank or de-group one or more flip-flops in the one or more MBFFs into one or more flip-flops not part of the one or more MBFFs. In some embodiments, the banking or de-banking engine are part of other portions of IC design and manufacturing flow **300**. For example, in some embodiments, the banking or de-banking engine are part of IC design operation **310**. In some embodiments, the banking engine or de-banking engine are aware of the analysis performed by MBFF power analysis **350**.

(86) In some embodiments, MBFF power analysis **350** is performed at the cell or block level. For example, MBFF power analysis **350** is performed for a region of the IC, instead of the whole IC. In at least one embodiment, a region of the IC includes a standard cell. In at least one embodiment, a region includes a group or a block of standard cells of the same or similar type or function, e.g., a block or group of MBFF cells, or a block or group of flip-flop cells, or the like. In at least one embodiment, a region includes a group or block of standard cells of different types or functions coupled together provide a function or module for the IC, e.g., a communication interface.

(87) Other arrangements for dividing the IC into multiple regions for the MBFF power analysis **350** are within the scopes of various embodiments. For simplicity, in the detailed description of one or more embodiments herein, a standard cell or cell is used as an example of a region of an IC.

(88) In some embodiments, MBFF power analysis **350** is performed at an early design stage. In at least one embodiment, MBFF power analysis **350** is performed between cell placement operation **324** and clock tree synthesis operation **326**, as indicated by arrow **360** in FIG. **3**. In at least one embodiment, MBFF power analysis **350** is performed between clock tree synthesis operation **326** and routing operation **328**, as indicated by arrow **370** in FIG. **3**. In at least one embodiment, MBFF power analysis **350** is performed between routing operation **328** and sign-off operation **330**, as indicated at arrow **380** by FIG. **3**. In at least one embodiment, MBFF power analysis **350** is performed multiple times in IC design flow **300**, for example, at two or more stages indicated by arrows **360**, **370**, **380**. In at least one embodiment, sign-off operation **330** still includes an MBFF power analysis **350** at the system level despite the fact that MBFF power analysis **350** has been performed earlier after one or more of cell placement **324**, clock tree synthesis operation **326** or routing operation **328**. In at least one embodiment, MBFF power analysis **350** is performed to determine power optimization of at least one standard cell stored in one or more cell libraries **325**, as indicated by arrow **390** in FIG. **3**. In some embodiments, cell library **325** includes an MBFF library **720** (shown in FIG. **7**) that includes a number of MBFF standard cells.

(89) As described herein, in some embodiments, MBFF power analysis **350** is performed to optimize the power consumed by one or more flip-flops, such as MBFF **100** or circuit **200A** or **200B** of FIG. **2A** or **2B**, while also passing a timing analysis of the flip-flops in the integrated

circuit. In some embodiments, an example of MBFF power analysis **350** is shown as method **400** of FIG. **4**. In some embodiments, MBFF power analysis **350** includes replacing a group of flip-flops with a banked MBFF having a plurality of flip-flops having different current driving abilities. (90) In some embodiments, by having different driving current capabilities, the corresponding flip-flops are able to switch states fast enough in order to pass timing tests or timing violations, but also do not consume additional power by being overdesigned by having a driving current capability more than needed in order to pass the timing tests or timing violations. Thus, one or more embodiments of the present disclosure are configured to optimize power and pass timing violations during the same operation, resulting in MBFFs that consume less power and occupy less area than other approaches. Furthermore, in some embodiments, one or more embodiments of the present disclosure are configured to optimize power and pass timing violations during the same operation resulting in a design and manufacturing flow **300** that includes less steps than other approaches that always de-bank flip-flops that are part of an MBFF that does not pass one or more timing violations.

(91) FIG. **4** is a flow chart of a method **400**, in accordance with some embodiments.

(92) In at least one embodiment, method **400** corresponds to MBFF power analysis **350**, and is performed in whole or in part by a processor, such as processor **702** of FIG. **7**.

(93) It is understood that additional operations may be performed before, during, and/or after the method **400** depicted in FIG. **4**, and that some other processes may only be briefly described herein. It is understood that method **400** utilizes features of one or more of MBFF **100** of FIG. **1**, or circuits **200A-200B** of corresponding FIGS. **2A-2B**. While the details of method **400** are described with respect to circuit elements, such as flip-flops or MBFFs, it is understood that the details of method **400** is applicable to circuit elements within a corresponding layout diagram, such as layout diagram **500A-500B** of FIGS. **5A-5B** or layout diagrams **600A-600D** of corresponding FIGS. **6A-6D**. For example, it is understood that method **400** utilizes features of one or more of layout diagrams **500A-500B** of corresponding FIGS. **5A-5B**, or layout diagrams **600A-600D** of corresponding FIGS. **6A-6D**.

(94) In operation **402** of method **400**, a set of flip-flops **501A** (FIG. **5A**) are banked or grouped into at least one MBFF (e.g., MBFF **501B** of FIG. **5B**) having a same driving current capability. In some embodiments, whether a set of flip-flops are banked or grouped into at least one MBFF **100** depends upon a number of criteria. In some embodiments, the criteria includes the proximity or distance between the flip-flops, whether the flip-flops share a same clock signal CLK, a same set signal S1, a same reset signal R1, a same scan enable signal SE1 or the like, or share a same corresponding pin. In other words, in these embodiments, flip-flops that are located from each other by a distance that will result in delays in at least the clock signal CLK, the set signal S1, the reset signal R1, or the scan enable signal SE1 will likely cause timing violations (described below in operation **406**) which will prevent the set of flip-flops from being banked or grouped into at least one MBFF **501B**. In some embodiments, operation **402** of method will attempt to bank or group as many flip-flops as possible into the at least one MBFF provided no timing violations occur. In other words, in some embodiments, the timing violations are at least one upper limit on the number of flip-flops that can be banked or grouped together. In some embodiments, the at least one MBFF of method **400** includes two or more flip-flops or corresponding bits. Other numbers of bits or corresponding flip-flops in MBFF are within the scope of the present disclosure.

(95) In some embodiments, the set of flip-flops **501A** includes flip-flops **502a**, **502b**, **502c** and **502d** (FIG. **5A**). Other numbers of flip-flops of the set of flip-flops **501A** are within the scope of the present disclosure. In some embodiments, the MBFF **501B** includes flip-flops **502'**, **504'**, **506'** and **508'** (FIG. **5B**). Other numbers of flip-flops in MBFF **501B** are within the scope of the present disclosure.

(96) In some embodiments, each of the flip-flops in the set of flip-flops **501A** or MBFF **501B** has the first driving current capability FF1. Other driving current capability for the set of flip-flops

501A or **MBFF 501B** is within the scope of the present disclosure. For example, in some embodiments, each of flip-flops in the set of flip-flops **501A** or **MBFF 501B** has the second driving current capability **FF0**.

(97) In operation **404** of method **400**, a timing analysis of each flip-flop **502'**, **504'**, **506'**, **508'** in at least multi-bit flip-flop **501B** is performed. In some embodiments, operation **404** includes operation **406**. In some embodiments, the timing analysis of operation **404** includes determining if a timing event occurs outside of a clock period of a clock signal **CLK**. For example, in some embodiments, a timing violation occurs if one or more of flip-flops **502'**, **504'**, **506'** and **508'** do not transition from a first state to a second state within a clock period of clock signal **CLK**.

(98) In operation **406** of method **400**, a determination is made if a timing violation occurs for each flip-flop **502'**, **504'**, **506'**, **508'** in at least multi-bit flip-flop **501B**. In some embodiments, a timing violation occurs for a negative slack on a path having a flip-flop. In some embodiments, slack is a difference between an actual time and a desired time for a timing path having a flip-flop of **MBFF 501B**. Thus, a negative slack occurs when the actual time is greater than the desired time for the corresponding timing path. Similarly, a positive slack occurs when the actual time is less than the desired time for the corresponding timing path. In some embodiments, if a positive slack is determined in operation **406**, then no timing violation occurs for the corresponding flip-flop in the corresponding timing path, the result of operation **406** is a “no”, and method **400** proceeds to operation **412**. In some embodiments, if a negative slack is determined in operation **406**, then a timing violation occurs for the corresponding flip-flop in the corresponding timing path, the result of operation **406** is a “yes”, and method **400** proceeds to operation **410**. In some embodiments, at least operation **404** or **406** is performed by a static timing analysis EDA tool.

(99) In operation **408** of method **400**, a set of flip-flops in at least the multi-bit flip-flop **501B** are downsized or upsized. For example, as shown in FIG. 6B, set of flip-flops **612** of **MBFF 601B** are downsized. For example, as shown in FIG. 6D, set of flip-flops **610** of **MBFF 601D** are upsized. In some embodiments, operation **408** includes at least operation **410**, operation **412**, operation **414** or operation **416**.

(100) In operation **410** of method **400**, a set of flip-flops **610** in at least the multi-bit flip-flop **501B** are upsized (e.g., shown in **MBFF 601D**). In some embodiments, if a timing violation occurs in a path, for example the determination of operation **406** is “yes”, then method **400** will upsize the flip-flops that had a timing violation in attempting to remove the timing violation. In some embodiments, upsizing the flip-flops that had a timing violation includes increasing the driving current capability of the corresponding flip-flops thereby increasing the switching speed of the corresponding flip-flops. For example, as shown in FIG. 6D, set of flip-flops **610** of **MBFF 601D** are upsized and have a third driving current capability **FF2**, while the driving current capability of set of flip-flops **506'** and **508'** are maintained at their previous driving current capability (e.g., **FF1**).

(101) After operation **410**, method **400** returns to operation **404** to perform additional timing analysis. In some embodiments, the additional timing analysis performed in operation **404** is to determine if the upsized flip-flops still have timing violations or new timing violations.

(102) In operation **412** of method **400**, a determination is made if flip-flops can be downsized. In some embodiments, if the determination of operation **412** is “yes”, then method **400** proceeds to operation **414** where a set of flip-flops **512** will be downsized. In some embodiments, if the determination of operation **412** is “no”, then method **400** proceeds to operation **416**. In some embodiments, if a previous set of flip-flops were upsized in operation **410**, and then operation **406** determined that the upsized flip-flops path had no timing violation, then the result of operation **412** for the upsized flip-flops is “no,” since the upsized flip-flops path should not be downsized as it was just previously upsized and did not include a timing violation.

(103) In operation **414** of method **400**, a set of flip-flops **512** in at least the multi-bit flip-flop **501B** are downsized (e.g., shown as the set of flip-flops **612** in **MBFF 601B** of FIG. 6B). In some embodiments, if no timing violation occurs in a path, for example the determination of operation

406 is “no”, then operation **414** of method **400** will downsize the flip-flops that did not have a timing violation in attempting to save power. In some embodiments, downsizing the flip-flops that did not have a timing violation includes decreasing the driving current capability of the corresponding flip-flops thereby decreasing the switching speed of the corresponding flip-flops thereby reducing the dynamic power consumed by the corresponding flip-flops. For example, as shown in FIG. **6B**, set of flip-flops **612** of MBFF **601B** are downsized and have the second driving current capability **FF0**, while the driving current capability of set of flip-flops **502'** and **504'** are maintained at their previous driving current capability (e.g., **FF1**).

(104) After operation **414**, method **400** returns to operation **404** to perform additional timing analysis. In some embodiments, the additional timing analysis performed in operation **404** is to determine if the downsized flip-flops have timing violations. In some embodiments, if the downsized flip-flops now have timing violations, then the downsized flip-flops are returned to their previous size (e.g., with their previous driving current capability).

(105) In operation **416** of method **400**, a determination is made if more or additional flip-flops are available in the at least one MBFF (e.g., MBFF **501B** of FIG. **5B**) or the banked set of flip-flops. In some embodiments, if the determination of operation **416** is “yes”, then method **400** returns to operation **404** where additional timing analysis is performed on the additional flip-flops that are banked in the set of flip-flops **512** or additional timing analysis is performed for other critical paths having a corresponding flip-flop in the at least one MBFF. In some embodiments, if the determination of operation **416** is “no”, then method **400** proceeds to operation **418**.

(106) In operation **418** of method **400**, a determination is made if more flip-flops are available in the layout diagram, such as layout diagram **500A**, that were not part of the at least one MBFF. In some embodiments, operation **418** further includes a determination if additional flip-flops were added to the layout diagram, such as layout diagram **500A**. In some embodiments, if the determination of operation **418** is “yes”, then method **400** proceeds to operation **402** where method **400** will attempt to bank the additional flip-flops into the at least one MBFF or into other MBFFs. In some embodiments, if the determination of operation **418** is “no”, then method **400** proceeds to operation **420**.

(107) In operation **420**, method **400** is in an idle state. In some embodiments, the idle state corresponds to method **400** waiting for an update from one of the parameters of operations **402-420**. In some embodiments, method **400** remains in the idle state until additional flip-flops are added to the layout diagram, such as layout diagram **500A**. In some embodiments, if additional flip-flops are added to the layout diagram, such as layout diagram **500A**, then operation **420** may return (not shown in FIG. **4**) to operation **418**. In some embodiments, the idle state of operation **420** may include the end of method **400**.

(108) In some embodiments, at least operation **402**, **404**, **406**, **408**, **410**, **412**, **414**, **416**, **418** or **420** is performed by an EDA tool, such as system **700** of FIG. **7**.

(109) In some embodiments, at least one method(s), such as method **300** or **400** discussed above, is performed in whole or in part by at least one EDA system, including system **700**. In some embodiments, an EDA system is usable as part of a design house of an IC manufacturing system **800** of FIG. **8**.

(110) In some embodiments, one or more of the operations of method **400** is not performed. While method **400** was described above with reference to FIGS. **1**, **2A-2B**, it is understood that method **400** utilizes the features of one or more of FIGS. **5A-5B** & **6A-6D**. In some embodiments, other operations of method **400** would be performed consistent with the description and operation of layout diagrams **500A-500B** & **600A-600D** of corresponding FIGS. **5A-5B** & **6A-6D**.

(111) Furthermore, in some embodiments, one or more embodiments of the present disclosure are configured to optimize power and pass timing violations during the same operation resulting in a method **400** that includes less steps than other approaches that always de-bank flip-flops that are part of an MBFF that does not pass one or more timing violations.

Examples of Method 400

(112) FIGS. 5A-5B are schematic views of layout diagrams 500A-500B of flip-flops before and after modifications, in accordance with some embodiments. The modifications are made as part of method 400 described with respect to FIG. 4.

(113) FIG. 5A is a schematic view of a layout diagram 500A of a set of flip-flops, e.g., set of flip-flops 501A, before execution of operation 402 of method 400 of FIG. 4.

(114) The set of flip-flops 501A includes flip-flops 502a, 502b, 502c and 502d. Other numbers of flip-flops of the set of flip-flops 501A are within the scope of the present disclosure. Each of the flip-flops 502a, 502b, 502c and 502d has a corresponding inverter 520a, 520b, 520c and 520d, and a corresponding inverter 522a, 522b, 522c and 522d. In some embodiments, each of the flip-flops in the set of flip-flops 501A has the first driving current capability FF1. Other driving current capability for the set of flip-flops 501A is within the scope of the present disclosure. For example, in some embodiments, each of flip-flops in the set of flip-flops 501A has the second driving current capability FF0.

(115) FIG. 5B is a schematic view of a layout diagram 500B of a multi-bit flip-flop, e.g., MBFF 501B, after execution of operation 402 of method 400 of FIG. 4.

(116) MBFF 501B includes flip-flops 502', 504', 506' and 508' and inverters 520a' and 522a'. Other numbers of flip-flops in MBFF 501B are within the scope of the present disclosure. In some embodiments, each of the flip-flops in MBFF 501B has a same driving current capability or the first driving current capability FF1. Other driving current capability for MBFF 501B is within the scope of the present disclosure. For example, in some embodiments, each of flip-flops in MBFF 501B has the second driving current capability FF0.

(117) FIGS. 6A-6D are schematic views of layout diagrams 600A-600D of flip-flops before and after modifications, in accordance with some embodiments. The modifications are made as part of method 400 described with respect to FIG. 4.

(118) FIG. 6A is a schematic view of a layout diagram 600A of a multi-bit flip-flop, e.g., MBFF 501B, before execution of operation 414 of method 400 of FIG. 4.

(119) MBFF 501B includes flip-flops 502', 504', 506' and 508'. In some embodiments, each of the flip-flops in MBFF 501B has a same driving current capability or the first driving current capability FF1. Set of flip-flops 512 includes flip-flops 506' and 508'. Set of flip-flops 512 in multi-bit flip-flop 501B are to be downsized in operation 414 of method 400.

(120) FIG. 6B is a schematic view of a layout diagram 600B of a multi-bit flip-flop, e.g., MBFF 601B, after execution of operation 414 of method 400 of FIG. 4.

(121) MBFF 601B includes flip-flops 502', 504', 606 and 608. Set of flip-flops 612 includes flip-flops 606 and 608. Set of flip-flops 612 were downsized in operation 414 of method 400. In some embodiments, flip-flops 502' and 504' have a same driving current capability or the first driving current capability FF1. In some embodiments, flip-flops 606 and 608 have a same driving current capability or the second driving current capability FF0.

(122) FIG. 6C is a schematic view of a layout diagram 600C of a multi-bit flip-flop, e.g., MBFF 501B, before execution of operation 410 of method 400 of FIG. 4.

(123) MBFF 501B includes flip-flops 502', 504', 506' and 508' and inverters 520a' and 522a'. In some embodiments, each of the flip-flops in MBFF 501B has a same driving current capability or the first driving current capability FF1. Set of flip-flops 510 includes flip-flops 502' and 504'. Set of flip-flops 510 in multi-bit flip-flop 501B are to be upsized in operation 410 of method 400.

(124) FIG. 6D is a schematic view of a layout diagram 600D of a multi-bit flip-flop, e.g., MBFF 601D, after execution of operation 410 of method 400 of FIG. 4.

(125) MBFF 601D includes flip-flops 602, 604, 506' and 508'. Set of flip-flops 612 includes flip-flops 606 and 608. Set of flip-flops 610 were upsized in operation 410 of method 400. In some embodiments, the set of flip-flops 610 had timing violations in operation 406, and therefore were upsized in order to overcome the timing violations of operation 406, and also optimized the total

clock dynamic power consumption of MBFF **601D**.

(126) In some embodiments, flip-flops **602** and **604** have a same driving current capability or the third driving current capability **FF2**. In some embodiments, the third driving current capability **FF2** is greater than the first driving current capability **FF1** and the second driving current capability **FF0**.

(127) In some embodiments, by having different driving current capabilities, the corresponding flip-flops of MBFF **601B** or **601D** are able to switch states fast enough in order to pass timing tests or timing violations, but also do not consume additional power by being overdesigned by having a driving current capability more than needed in order to pass the timing tests or timing violations. Thus, one or more embodiments of the present disclosure are configured to optimize power and pass timing violations during the same operation, resulting in MBFF **601B** or **601D** consuming less power and occupying less area than other approaches.

(128) FIG. **7** is a schematic view of a system **700** for designing an IC layout design, in accordance with some embodiments. In some embodiments, system **700** is at least a part of an EDA system. In some embodiments, system **700** includes an automated placement and routing (APR) system. In some embodiments, system **700** generates or places one or more IC layout designs described herein. In some embodiments, the IC layout designs of FIG. **7** includes at least layout diagrams **500A-500B** of corresponding FIGS. **5A-5B** and layout diagrams **600A-600D** of corresponding FIGS. **6A-6D**.

(129) System **700** includes a hardware processor **702** and a non-transitory, computer readable storage medium **704** (shown in FIG. **7** as “Memory **704**”) encoded with, i.e., storing, the computer program code **706**, i.e., a set of executable instructions. Computer readable storage medium **704** is configured for interfacing with manufacturing machines for producing the integrated circuit. The processor **702** is electrically coupled to the computer readable storage medium **704** via a bus **708**. The processor **702** is also electrically coupled to an I/O interface **710** by bus **708**. A network interface **712** is also electrically connected to the processor **702** via bus **708**. Network interface **712** is connected to a network **714**, so that processor **702** and computer readable storage medium **704** are capable of connecting to external elements via network **714**. The processor **702** is configured to execute the computer program code **706** encoded in the computer readable storage medium **704** in order to cause system **700** to be usable for performing a portion or all of the operations as described in method **300** or **400**.

(130) In some embodiments, the processor **702** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(131) In some embodiments, the computer readable storage medium **704** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the computer readable storage medium **704** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments using optical disks, the computer readable storage medium **704** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(132) In some embodiments, the storage medium **704** stores the computer program code **706** configured to cause system **700** to perform method **300** or **400**. In some embodiments, the storage medium **704** also stores information needed for performing method **300** or **400** as well as information generated during performing method **300** or **400**, such as MBFF engine **716**, user interface **718**, MBFF library **720**, cell library **722**, MBFF power analysis **724** and layout diagram **726**, and/or a set of executable instructions to perform the operation of method **300** or **400**. In some embodiments, layout diagram **726** comprises one or more of layout diagrams **500A-500B** of corresponding FIGS. **5A-5B** and layout diagrams **600A-600D** of corresponding FIGS. **6A-6D**.

(133) In some embodiments, the storage medium **704** stores instructions (e.g., computer program

code **706**) for interfacing with manufacturing machines. The instructions (e.g., computer program code **706**) enable processor **702** to generate manufacturing instructions readable by the manufacturing machines to effectively implement method **300** or **400** during a manufacturing process.

(134) System **700** includes I/O interface **710**. I/O interface **710** is coupled to external circuitry. In some embodiments, I/O interface **710** includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to processor **702**.

(135) System **700** also includes network interface **712** coupled to the processor **702**. Network interface **712** allows system **700** to communicate with network **714**, to which one or more other computer systems are connected. Network interface **712** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1394. In some embodiments, method **300** or **400** is implemented in two or more systems **700**, and information such as MBFF engine, user interface, MBFF library, cell library, MBFF power analysis and layout diagram are exchanged between different systems **700** by network **714**.

(136) System **700** is configured to receive information related to a layout diagram through I/O interface **710** or network interface **712**. The information is transferred to processor **702** by bus **708** to determine a layout diagram for producing MBFF **100**, circuit **200A** or **200B**. The layout diagram is then stored in computer readable medium **704** as layout design **726**. System **700** is configured to receive information related to an MBFF engine through I/O interface **710** or network interface **712**. The information is stored in computer readable medium **704** as MBFF engine **716**. System **700** is configured to receive information related to a user interface through I/O interface **710** or network interface **712**. The information is stored in computer readable medium **704** as user interface **718**. System **700** is configured to receive information related to a MBFF library through I/O interface **710** or network interface **712**. The information is stored in computer readable medium **704** as MBFF library **720**. System **700** is configured to receive information related to a cell library through I/O interface **710** or network interface **712**. The information is stored in computer readable medium **704** as cell library **722**. System **700** is configured to receive information related to a MBFF power analysis library through I/O interface **710** or network interface **712**. The information is stored in computer readable medium **704** as MBFF power analysis **724**.

(137) In some embodiments, method **300** or **400** is implemented as a standalone software application for execution by a processor. In some embodiments, method **300** or **400** is implemented as a software application that is a part of an additional software application. In some embodiments, method **300** or **400** is implemented as a plug-in to a software application. In some embodiments, method **300** or **400** is implemented as a software application that is a portion of an EDA tool. In some embodiments, method **300** or **400** is implemented as a software application that is used by an EDA tool. In some embodiments, the EDA tool is used to generate a layout diagram of the integrated circuit device. In some embodiments, the layout is stored on a non-transitory computer readable medium. In some embodiments, the layout is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool. In some embodiments, the layout is generated based on a netlist which is created based on the schematic design. In some embodiments, method **300** or **400** is implemented by a manufacturing device to manufacture an integrated circuit using a set of masks manufactured based on one or more layout designs generated by system **700**. In some embodiments, system **700** a manufacturing device to manufacture an integrated circuit using a set of masks manufactured based on one or more layout designs of the present disclosure. In some embodiments, system **700** of FIG. 7 generates layout designs of an integrated circuit that are smaller than other approaches. In some embodiments, system **700** of FIG. 7 generates layout designs of integrated circuit structure that occupy less area and provide better routing resources than other approaches.

(138) FIG. 8 is a block diagram of an integrated circuit (IC) manufacturing system **800**, and an IC

manufacturing flow associated therewith, in accordance with at least one embodiment of the present disclosure.

(139) In FIG. 8, IC manufacturing system **800** includes entities, such as a design house **820**, a mask house **830**, and an IC manufacturer/fabricator (“fab”) **840**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **860**. The entities in system **800** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **820**, mask house **830**, and IC fab **840** is owned by a single larger company. In some embodiments, two or more of design house **820**, mask house **830**, and IC fab **840** coexist in a common facility and use common resources.

(140) Design house (or design team) **820** generates an IC design layout **822**. IC design layout **822** includes various geometrical patterns designed for an IC device **860**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **860** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout **822** includes various IC features, such as an active region, gate electrode, source electrode and drain electrode, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **820** implements a proper design procedure to form IC design layout **822**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout **822** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout **822** can be expressed in a GDSII file format or DFII file format.

(141) Mask house **830** includes data preparation **852** and mask fabrication **844**. Mask house **830** uses IC design layout **822** to manufacture one or more masks to be used for fabricating the various layers of IC device **860** according to IC design layout **822**. Mask house **830** performs mask data preparation **852**, where IC design layout **822** is translated into a representative data file (“RDF”). Mask data preparation **852** provides the RDF to mask fabrication **844**. Mask fabrication **844** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) or a semiconductor wafer. The design layout is manipulated by mask data preparation **852** to comply with particular characteristics of the mask writer and/or requirements of IC fab **840**. In FIG. 8, mask data preparation **852** and mask fabrication **844** are illustrated as separate elements. In some embodiments, mask data preparation **852** and mask fabrication **844** can be collectively referred to as mask data preparation.

(142) In some embodiments, mask data preparation **852** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout **822**. In some embodiments, mask data preparation **852** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

(143) In some embodiments, mask data preparation **852** includes a mask rule checker (MRC) that checks the IC design layout that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some

embodiments, the MRC modifies the IC design layout to compensate for limitations during mask fabrication **844**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

(144) In some embodiments, mask data preparation **852** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **840** to fabricate IC device **860**. LPC simulates this processing based on IC design layout **822** to create a simulated manufactured device, such as IC device **860**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout **822**.

(145) It should be understood that the above description of mask data preparation **852** has been simplified for the purposes of clarity. In some embodiments, data preparation **852** includes additional features such as a logic operation (LOP) to modify the IC design layout according to manufacturing rules. Additionally, the processes applied to IC design layout **822** during data preparation **852** may be executed in a variety of different orders.

(146) After mask data preparation **852** and during mask fabrication **844**, a mask or a group of masks are fabricated based on the modified IC design layout. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) based on the modified IC design layout. The mask can be formed in various technologies. In some embodiments, the mask is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the mask. In another example, the mask is formed using a phase shift technology. In the phase shift mask (PSM), various features in the pattern formed on the mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **844** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in the semiconductor wafer, in an etching process to form various etching regions in the semiconductor wafer, and/or in other suitable processes.

(147) IC fab **840** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **840** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

(148) IC fab **840** uses the mask (or masks) fabricated by mask house **830** to fabricate IC device **860**. Thus, IC fab **840** at least indirectly uses IC design layout **822** to fabricate IC device **860**. In some embodiments, a semiconductor wafer **842** is fabricated by IC fab **840** using the mask (or masks) to form IC device **860**. Semiconductor wafer **842** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

(149) Details regarding an integrated circuit (IC) manufacturing system (e.g., system 800 of FIG. 8), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20150278429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

(150) One aspect of this description relates to a multi-bit flip-flop. In some embodiments, the multi-bit flip-flop includes a first flip-flop having a first driving capability, the first flip-flop including a first set pin configured to receive a first set signal. In some embodiments, the multi-bit flip-flop further includes a second flip-flop having a second driving capability different from the first driving capability, the second flip-flop including a second set pin configured to receive the first set signal, and the first set pin and the second set pin are coupled together. In some embodiments, the first flip-flop and the second flip-flop are configured to share at least a first clock pin.

(151) Another aspect of this description relates to a method of forming an integrated circuit. In some embodiments, the method includes placing, by a processor, a standard cell layout of the integrated circuit, and manufacturing the integrated circuit based on the standard cell layout. In some embodiments, the placing of the standard cell layout includes determining if a first multi-bit flip-flop (MBFF) had a first timing violation, the first MBFF includes a first set of flip-flops, each flip-flop in the first set of flip-flops having a first driving current capability. In some embodiments, the placing of the standard cell layout further includes upsizing a second set of flip-flops in at least the first MBFF in response to determining that the first MBFF had the first timing violation, the second set of flip-flops having a second driving current capability different from the first driving current capability.

(152) Yet another aspect of this description relates to a system for designing an integrated circuit. In some embodiments, the system includes a non-transitory computer readable medium configured to store non-transitory instructions; and a processor coupled to the non-transitory computer readable medium. In some embodiments, the processor is configured to execute the non-transitory instructions including determining if a first multi-bit flip-flop (MBFF) had a first timing violation, the first MBFF includes a first set of flip-flops, each flip-flop in the first set of flip-flops having a first driving current capability. In some embodiments, the processor is configured to execute the non-transitory instructions further including upsizing a second set of flip-flops in at least the first MBFF in response to determining that the first MBFF had the first timing violation, the second set of flip-flops having a second driving current capability different from the first driving current capability.

(153) A number of embodiments have been described. It will nevertheless be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, various transistors being shown as a particular dopant type (e.g., N-type or P-type Metal Oxide Semiconductor (NMOS or PMOS)) are for illustration purposes. Embodiments of the disclosure are not limited to a particular type. Selecting different dopant types for a particular transistor is within the scope of various embodiments. The low or high logical value of various signals used in the above description is also for illustration. Various embodiments are not limited to a particular logical value when a signal is activated and/or deactivated. Selecting different logical values is within the scope of various embodiments. In various embodiments, a transistor functions as a switch. A switching circuit used in place of a transistor is within the scope of various embodiments. In various embodiments, a source of a transistor can be configured as a drain, and a drain can be configured as a source. As such, the term source and drain are used interchangeably. Various signals are generated by corresponding circuits, but, for simplicity, the circuits are not shown.

(154) Various figures show capacitive circuits using discrete capacitors for illustration. Equivalent circuitry may be used. For example, a capacitive device, circuitry or network (e.g., a combination of capacitors, capacitive elements, devices, circuitry, or the like) can be used in place of the discrete

capacitor. The above illustrations include exemplary operations or steps, but the steps are not necessarily performed in the order shown. Steps may be added, replaced, changed order, and/or eliminated as appropriate, in accordance with the spirit and scope of disclosed embodiments. (155) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A multi-bit flip-flop comprising: a first flip-flop having a first driving capability, the first flip-flop including a first set pin configured to receive a first set signal; and a second flip-flop having a second driving capability different from the first driving capability, the second flip-flop including a second set pin configured to receive the first set signal, and the first set pin and the second set pin are coupled together, wherein the first flip-flop and the second flip-flop are configured to share at least a first clock pin.
2. The multi-bit flip-flop of claim 1, further comprising: a third flip-flop having the first driving capability, the third flip-flop including a third set pin configured to receive the first set signal, and the third set pin being coupled to the first set pin and the second set pin, wherein the third flip-flop is configured to share the first clock pin with the first flip-flop and the second flip-flop.
3. The multi-bit flip-flop of claim 2, further comprising: a fourth flip-flop having the second driving capability, the fourth flip-flop including a fourth set pin configured to receive the first set signal, and the fourth set pin being coupled to the first set pin, the second set pin and the third set pin, wherein the fourth flip-flop is configured to share the first clock pin with the first flip-flop, the second flip-flop and the third flip-flop.
4. The multi-bit flip-flop of claim 1, wherein the first driving capability is based on the first flip-flop having a first number of fins; and the second driving capability is based on the second flip-flop having a second number of fins.
5. The multi-bit flip-flop of claim 4, wherein the first number of fins is different from the second number of fins.
6. The multi-bit flip-flop of claim 1, wherein the first driving capability is based on the first flip-flop having a first threshold voltage; and the second driving capability is based on the second flip-flop having a second threshold voltage.
7. The multi-bit flip-flop of claim 6, wherein the first threshold voltage is different from the second threshold voltage.
8. The multi-bit flip-flop of claim 1, wherein the first flip-flop further includes a first data input pin configured to receive a first data signal; and the second flip-flop further includes a second data input pin configured to receive a second data signal.
9. The multi-bit flip-flop of claim 8, wherein the first flip-flop further includes: a first reset pin configured to receive a first reset signal; and the second flip-flop further includes: a second reset pin configured to receive the first reset signal, and the first reset pin and the second reset pin are coupled together.
10. The multi-bit flip-flop of claim 1, wherein at least one of the first flip-flop or the second flip-flop is an SR flip-flop, a DQ flip-flop, JK flip-flop or a T-flip-flop.
11. A method of forming an integrated circuit, the method comprising: placing, by a processor, a standard cell of the integrated circuit, the placing of the standard cell comprising: determining if a

first multi-bit flip-flop (MBFF) had a first timing violation, the first MBFF includes a first set of flip-flops, each flip-flop in the first set of flip-flops having a first driving current capability; upsizing a second set of flip-flops in at least the first MBFF in response to determining that the first MBFF had the first timing violation, the second set of flip-flops having a second driving current capability different from the first driving current capability; and manufacturing the integrated circuit based on the standard cell.

12. The method of claim 11, wherein each flip-flop in the second set of flip-flops has a corresponding timing violation.

13. The method of claim 11, further comprising: determining if the first MBFF had a second timing violation; and upsizing a third set of flip-flops in the second set of flip-flops in at least the first MBFF in response to determining that the first MBFF had the second timing violation, wherein the third set of flip-flops have a third driving current capability different from the first driving current capability and the second driving current capability, and each flip-flop in the third set of flip-flops has a corresponding timing violation.

14. The method of claim 11, wherein each flip-flop in the first multi-bit flip-flop with the first driving current capability has a first number of fins, each flip-flop in the second set of flip-flops has a second number of fins, and the first number of fins is different from the second number of fins.

15. The method of claim 11, wherein each flip-flop in the first multi-bit flip-flop with the first driving current capability has a first threshold voltage, each flip-flop in the second set of flip-flops has a second threshold voltage, and the first threshold voltage is different from the second threshold voltage.

16. A system for designing an integrated circuit, the system comprises: a non-transitory computer readable medium configured to store non-transitory instructions; and a processor coupled to the non-transitory computer readable medium, wherein the processor is configured to execute the non-transitory instructions for: determining if a first multi-bit flip-flop (MBFF) had a first timing violation, the first MBFF includes a first set of flip-flops, each flip-flop in the first set of flip-flops having a first driving current capability; and upsizing a second set of flip-flops in at least the first MBFF in response to determining that the first MBFF had the first timing violation, the second set of flip-flops having a second driving current capability different from the first driving current capability.

17. The system of claim 16, wherein the processor is configured to execute the non-transitory instructions where the first driving current capability is based on the first set of flip-flops having a first number of fins, the second driving current capability is based on the second set of flip-flops having a second number of fins, and the first number of fins is different from the second number of fins.

18. The system of claim 16, wherein the processor is configured to execute the non-transitory instructions where the first driving current capability is based on the first set of flip-flops having a first threshold voltage, the second driving current capability is based on the second set of flip-flops having a second threshold voltage, and the first threshold voltage is different from the second threshold voltage.

19. The system of claim 16, wherein the processor is configured to execute the non-transitory instructions where each flip-flop in the second set of flip-flops has a corresponding first timing violation.

20. The system of claim 16, wherein the processor is configured to execute the non-transitory instructions further comprising: determining if the first MBFF had a second timing violation; and upsizing a third set of flip-flops in the second set of flip-flops in at least the first MBFF in response to determining that the first MBFF had the second timing violation, wherein the third set of flip-flops have a third driving current capability different from the first driving current capability and the second driving current capability, and each flip-flop in the third set of flip-flops has a corresponding second timing violation.

